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SUMMARY

AIML undergraduate at VIT Vellore with strong foundations in Data Structures, Machine Learning, and scalable system design. Ranked among the top students in the department with experience in full-stack development and building efficient, production-ready applications.

EDUCATION

Vellore Institute of Technology, Vellore

Bachelor of Technology (B.Tech), Computer Science (AI & ML) – Rank 1

CGPA: 9.68/10

Aug 2023 – Apr 2027

Sacred Heart Higher Secondary School, Sitapur (CBSE)

Secondary Education: 94.0%

Senior Secondary Education: 89.8%

Apr 2020 – July 2022

EXPERIENCE

Open Source Contributions

May 2026

Zerith DB

- Optimized query execution in ZerithDB by implementing cursor-based pagination and early-termination query scanning for large datasets.
- Reduced paginated query latency from **182ms to 0.5ms (~365x improvement)** on **10k-record benchmarks** while significantly lowering memory overhead.

Cubing Club, VIT Vellore

Mar 2024 – Present

Design Team Leader (June 2025 – Present)

- Led planning and execution of **5+ large-scale** technical and creative events during Aurora Week, managing logistics, timelines, and cross-functional coordination, attracting over **100+ participants**.
- Directed a **10+ member team** to design event frameworks, optimize execution workflows, and launch targeted promotional campaigns, increasing participation by **30%** (80 to 105+ attendees).

PROJECTS

Semantic Search Engine | SentenceTransformers, FAISS, Docker [\[GitHub\]](#)

Feb 2026

- Built a **semantic document search engine** using transformer embeddings and **FAISS vector search**, enabling **sub-100ms similarity retrieval** across high-dimensional embeddings.
- Designed a **cluster-aware semantic caching layer** using **Gaussian Mixture Model (fuzzy clustering)**, reducing lookup complexity from **O(N)** to **O(K)** for scalable query performance.

VastuZone | React, Node.js, MongoDB, Socket.IO, Razorpay, Cloudinary [\[Live\]](#)

Nov 2025

- Deployed a generative AI-powered Vastu consultation platform automating floor-plan analysis and personalized room-wise recommendation generation, reducing **manual user input time by 80%** (70 mins to <10s).
- Created a scalable backend with Firebase-JWT authentication, Zod validation, and REST + Socket.IO infrastructure, achieving secure RBAC enforcement and **15% lower request latency**.

Multi-Platform Fare Prediction System | FastAPI, scikit-learn, NumPy, HuggingFace, Render [\[Live\]](#)

Jan 2026

- Deployed a real-time ML pricing engine predicting ride fares across 4 platforms using gradient boosting with dynamic feature engineering (traffic, surge multiplier, weather APIs).
- Architected a FastAPI inference service **hosting(>200MB+)** ML models with **IP-scoped rate limiting and response caching**, achieving <150ms latency and **35%** compute reduction.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, Java, JavaScript

Machine Learning & Data Science: Scikit-learn, NumPy, Pandas, Transformers, Model Evaluation, Gradient Boosting

Frameworks & Technologies: FastAPI, Django REST Framework, Node.js, Express.js, React.js, JWT Authentication

Databases & DevOps: MongoDB, PostgreSQL, MySQL, Docker, Docker Compose, Git, AWS, Render, Vercel, Swagger

Core Computer Science: Data Structures & Algorithms, OOPS, DBMS, Operating Systems, Computer Networks

ACHIEVEMENTS

- Ranked **1st** in AIML Department among **250+ students**, demonstrating academic excellence.
- Solved 350+ Data Structures and Algorithms problems across LeetCode and other competitive coding platforms.
- Elected **Program Representative** for CSE-AIML, providing leadership and strategic coordination for a cohort of **250+ students**.